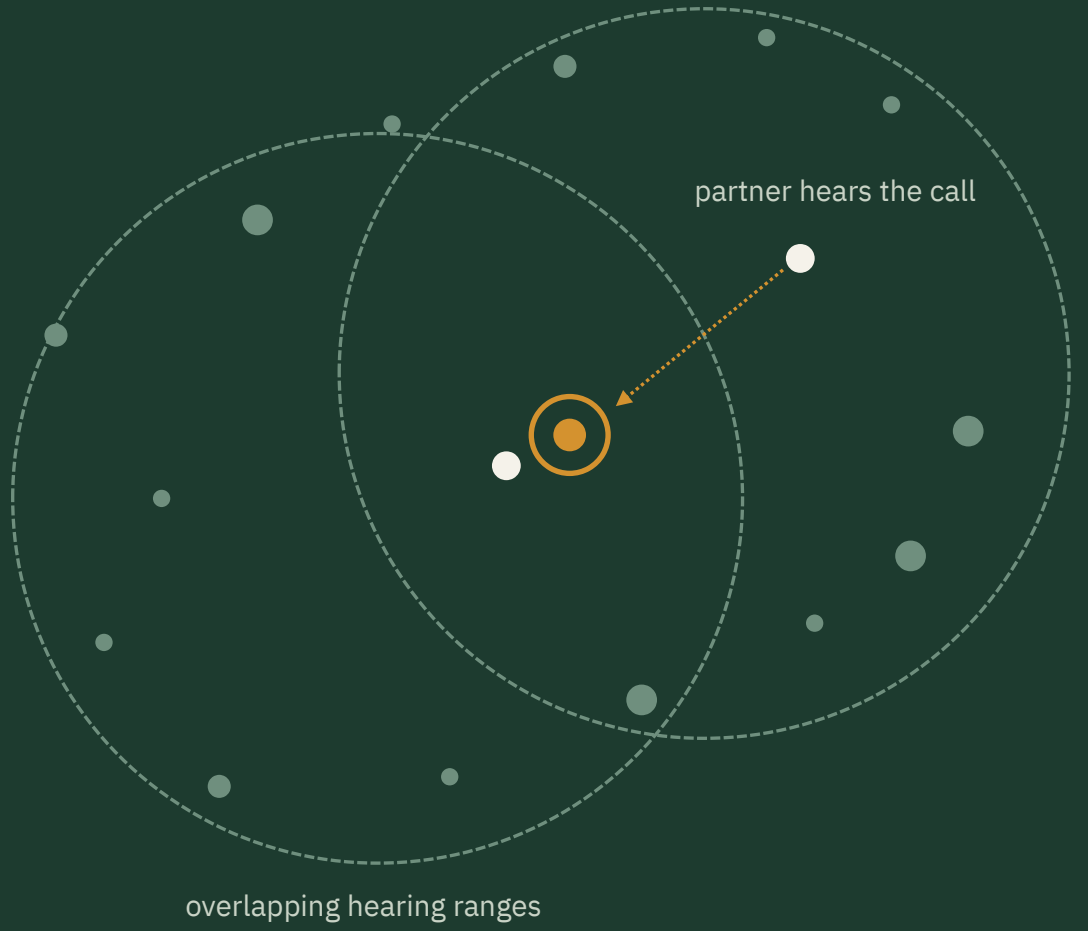
CANOPY

How talking, distance and honesty shape cooperative foraging



SCUDEM XI 2026 · Problem B

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Two animals forage together and can talk. What matters?



Distance

How far apart can they be and still hear each other?

Slides 15–16



Talking

Do they call at all, how often, and how far does a call carry?

Slide 11



Detail

“Food!” or “big tree, north, ripe for three more days”?

Slide 12



Accuracy

Honest mistakes, and deliberate lies

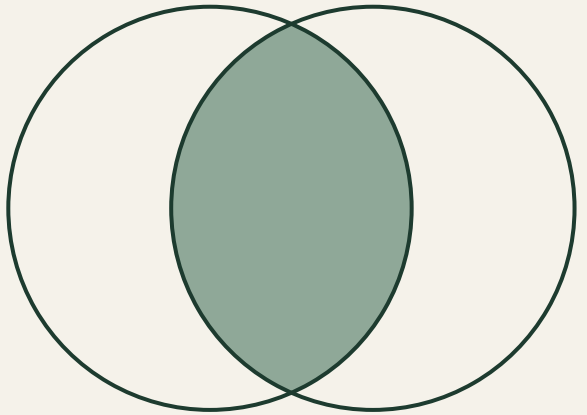
Slides 13, 14, 17

We measure success as pair intake: fruit both animals eat per day.

Then we rank the factors and ask how they trade off, e.g. distance against accuracy.

They split up to search, then pool what they know

Fission–fusion	Subgroups split and rejoin through the day
Range	500–4,500 m a day (site means 1.75–3.3 km), about 10 fruiting trees
Sharing	Monkeys new to a fruiting tree tend to arrive with ones who already know it
Seasons	Dry means scarcer, patchier fruit and larger, less overlapping core ranges



drawn to scale

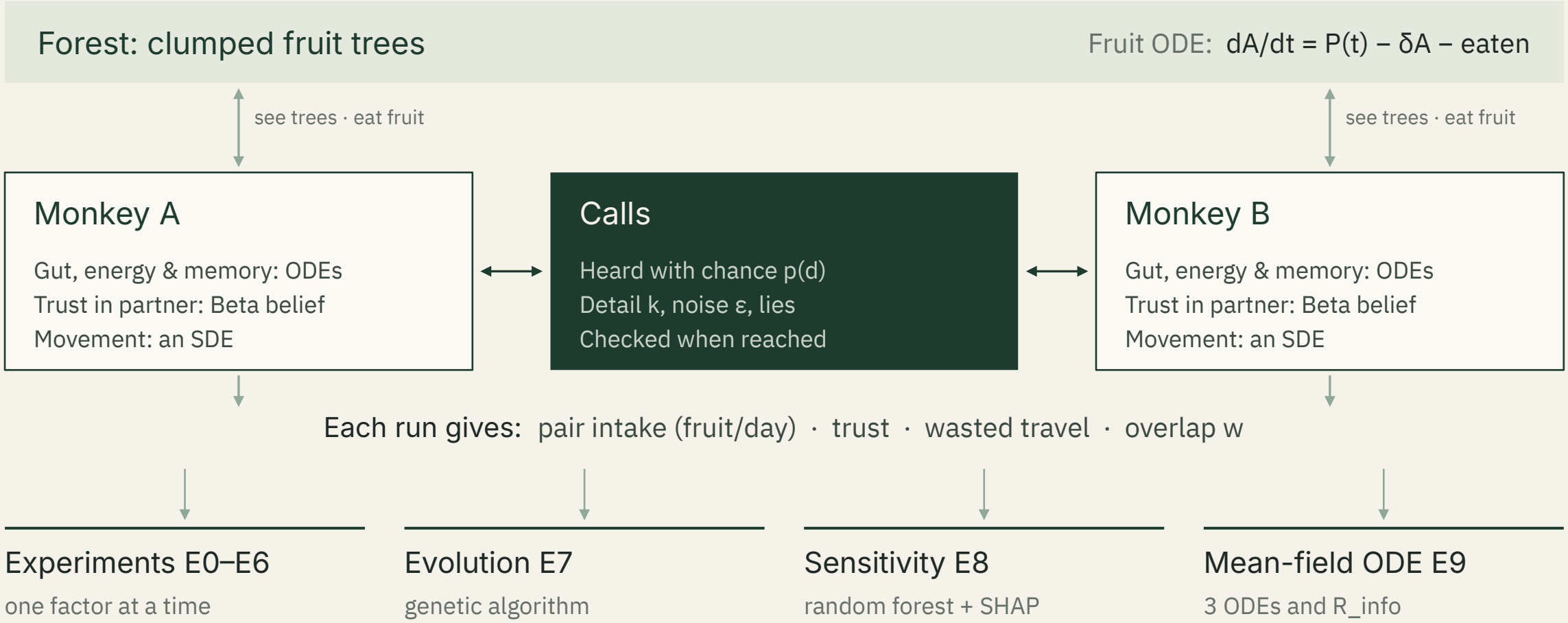
$w^* = 1/3$

Best overlap for a pair

Overlap = shared area ÷ combined area of two core ranges. The paper derives w* = 1/(n + 1): enough overlap to meet and share, little enough to know different trees.

Source: Ramos-Fernandez et al., npj Complexity 3, 1 (2026)

Agents that run on differential equations



Clumped trees whose fruit follows an ODE

Fruit on tree j

$$\frac{dA_j}{dt} = \underbrace{P_j(t)}_{\text{ripening}} - \underbrace{\delta A_j}_{\text{rot}} - \underbrace{\sum_i I_{ij}(t)}_{\text{eaten}}$$

Rot δ = 0.3/day: uneaten fruit halves in 2.3 days

Ripening over a window of W days

$$P(t) = \frac{2F}{W} \sin^2\left(\pi \frac{t - t_{on}}{W}\right)$$

Where: trees in Gaussian clumps. **How big:** lognormal crops, so a few trees are huge and worth sharing.

	Dry	Wet
Trees per 30 days	140	250
Clump spread	45 m	120 m
Median crop	300	500
Ripe for	3–8 days	12–30 days
Hyperabundant tree	none	40% of trees

Dry: scarce, patchy, short-lived fruit. Wet: plentiful and even, like the paper's *Brosimum* season.

It digests, eats and always chooses the most fruit per minute

Gut fills and empties

$$\frac{dG}{dt} = I(t) - k_d\, G$$

Half-life about 1 hour. Full gut = rest.

Eating saturates

$$I = \frac{r_{\max}\, A}{A + K} \cdot \frac{1}{1 + \gamma\, \mathbf{1}[\text{tree shared}]}$$

Holling type II; sharing a tree slows both.

Knowing when to leave

$$I < 0.8\, R$$

Marginal value theorem: leave when this tree pays less than average.

Choosing a tree

Expected fruit ÷ time to walk, eat and climb. Go if the best U beats R.

$$U_j = \frac{c_j\, g_j}{d_j / v + g_j / (0.6\, r_{\max}) + t_h}, \qquad g_j = \min(\hat{A}_j,\, G_{\max} - G)$$

States search → travel → feed → rest · regroup if contact is lost · home to the sleeping site at dusk

Movement is an SDE with a spacing spring

Where I want to go + spacing + noise

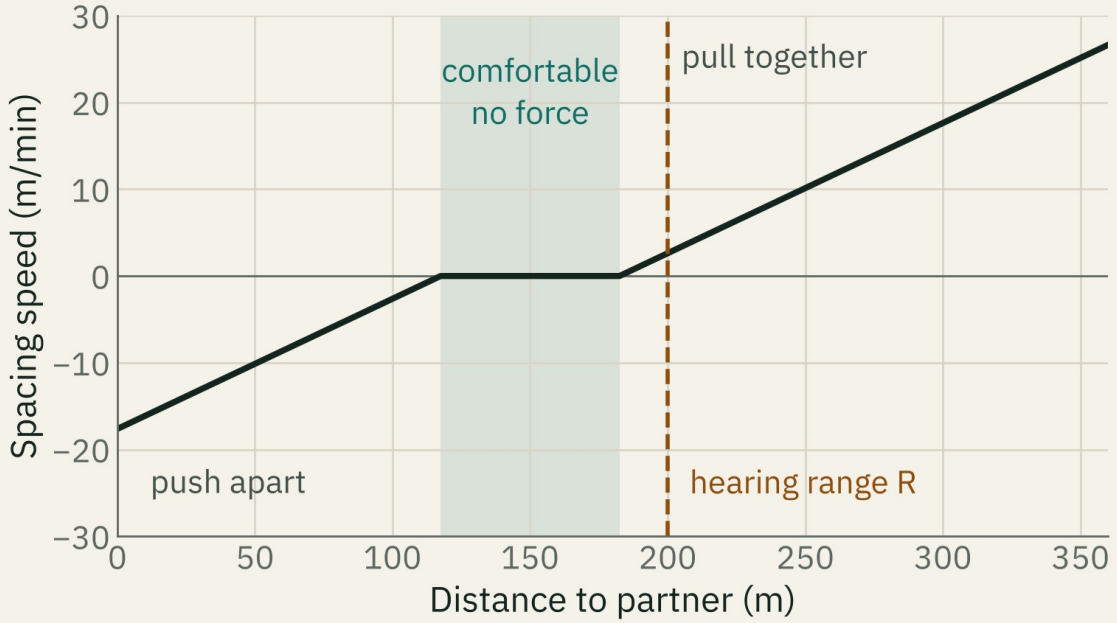
$$d\mathbf{x}_i = \left[\, v_0\, \hat{\mathbf{u}}_i + k_s\, s(d)\, \hat{\mathbf{d}}_i \,\right] dt + \sqrt{2D}\; d\mathbf{W}_i$$

Stepped each minute by Euler–Maruyama

Stay apart, stay in contact

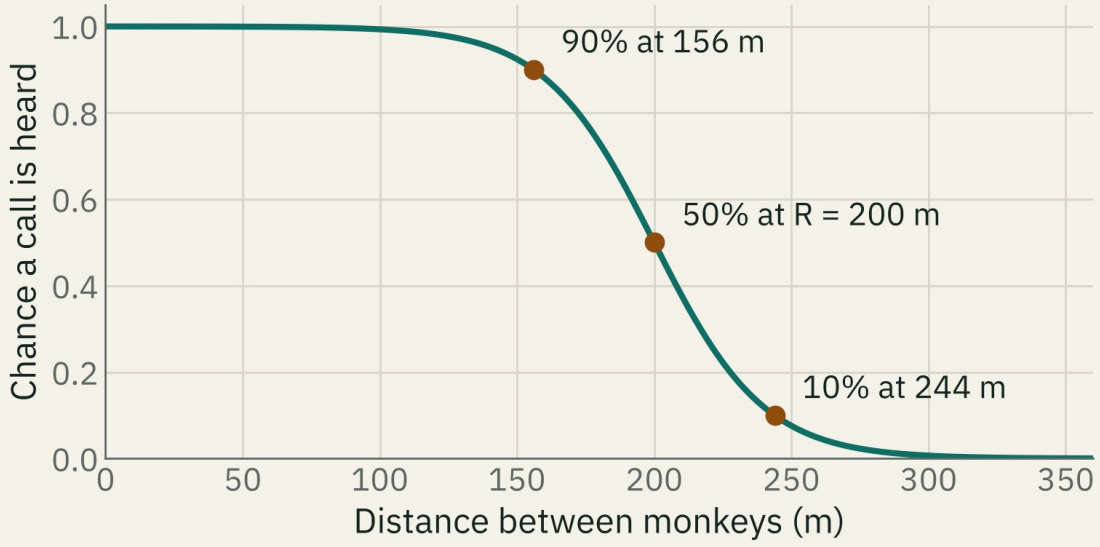
$$s(d) = \begin{cases} d - (d_{pref} + \Delta) & \text{too far: pull} \\ 0 & \text{comfortable} \\ d - (d_{pref} - \Delta) & \text{too close: push} \end{cases}$$

Searching: Lévy walk, step-length odds $\propto 1/\ell^2$ (half under 20 m, a few up to 400 m)



The comfortable band (118–182 m here, preferred spacing 0.75 R) has no force; the pull is toward where the partner was last heard.

Calls fade with distance; detail varies



$$p_{hear}(d) = \frac{1}{1 + e^{(d-R)/s}}$$

Chance a call is heard: logistic in distance, 50% at R = 200 m

Call only when the tree has 30+ fruit beyond your own need

k	The call says	Listener learns
0	nothing (no food calls)	the silent baseline
1	“food here”	where the caller is
2	+ size: small, medium, large	if the trip is worth it
3	+ 2 other trees; briefings when they meet	extra targets
4	+ when each tree runs out	how long the tip lasts

The listener goes if trust × reliability × fruit ÷ trip time beats what it is doing now. Every tip it follows is checked on arrival.

Tips are checked, and trust updates

Honest noise ε: each message field is wrong by chance. Lies: four tactics.

False alarm

points to an empty tree

Undersell

calls a big tree small

Misdirect

gives a spot 120–200 m off

Withhold

stays silent: never caught

Trust: a Beta belief, updated per check

$$\alpha_{n+1} = (1-\lambda)\,\alpha_n + \lambda\,\alpha_0 + o_n, \qquad T = \frac{\alpha}{\alpha+\beta}$$

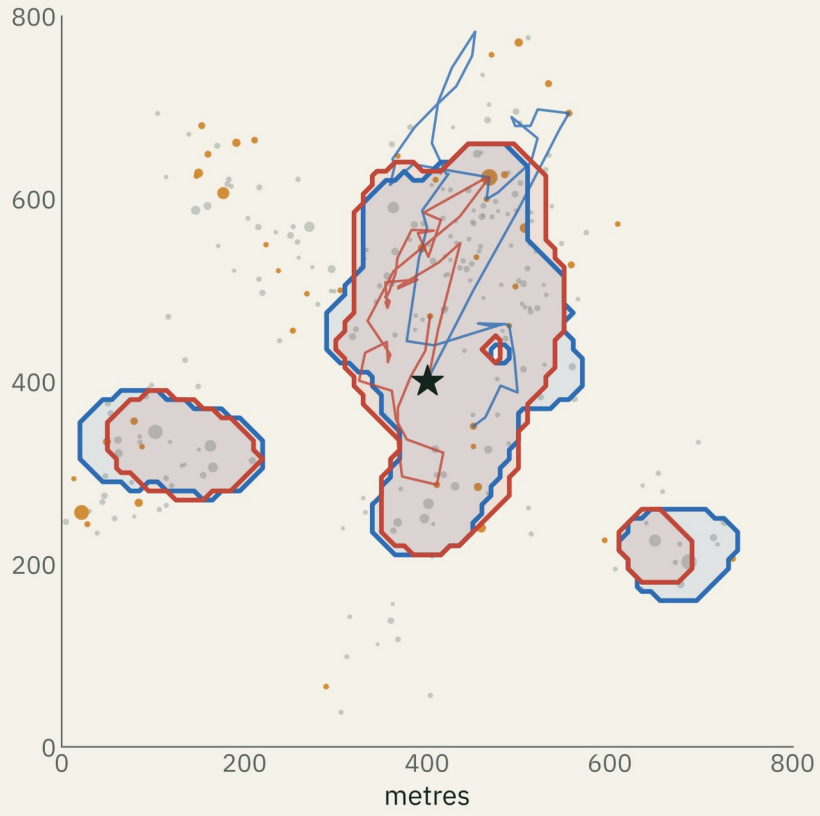
o = 1 if a tip checked out. It settles at

$$T^* = \frac{\alpha_0 + p/\lambda}{\alpha_0 + \beta_0 + 1/\lambda} = \frac{9 + 20p}{30}$$

0.97 if every tip checks out, 0.30 if none do

The listener cannot tell noise from a lie. Both lower trust, and low trust mutes even true calls.

Movement matches; overlap is too high



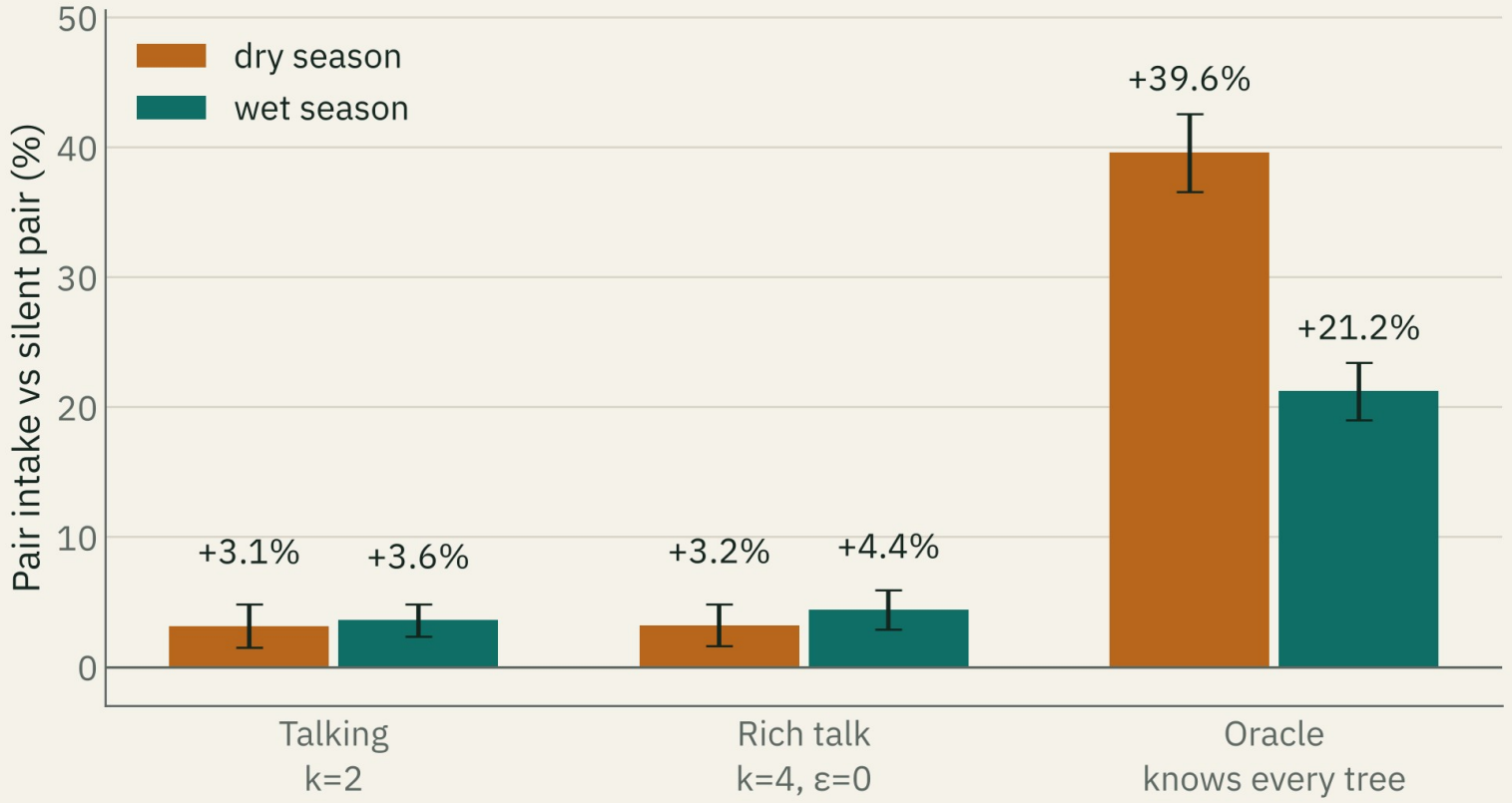
Dry season: outlines are 60% core ranges, lines are one day's paths, orange trees are ripe, star is the sleeping site.

Measure	Paper	Dry	Wet	
Daily path	means 1.75–3.3 km	2.9 km	2.1 km	match
Fruit trees used per day	about 10	5.9	9.5	miss (dry)
Core range larger in dry?	yes	11.1 ha	9.3 ha	match
Core-range overlap w	1/3 is optimal	0.72	0.64	miss

Matches: daily distance, and larger ranges in the dry season.

Misses: overlap is about twice the paper's optimum, since both monkeys start and end every day at one sleeping site; dry-season monkeys use fewer trees.

Talking helps, but only a little



20 forests per bar, compared forest by forest with the silent pair; 60-day seasons

+3%

more fruit for a pair that calls
both seasons, 95% intervals exclude zero

+21–40%

if they knew every tree
so calls capture only 8–20% of what perfect information is worth

A simple call gets most of the benefit



Wet

“Food here” alone gives the full gain.

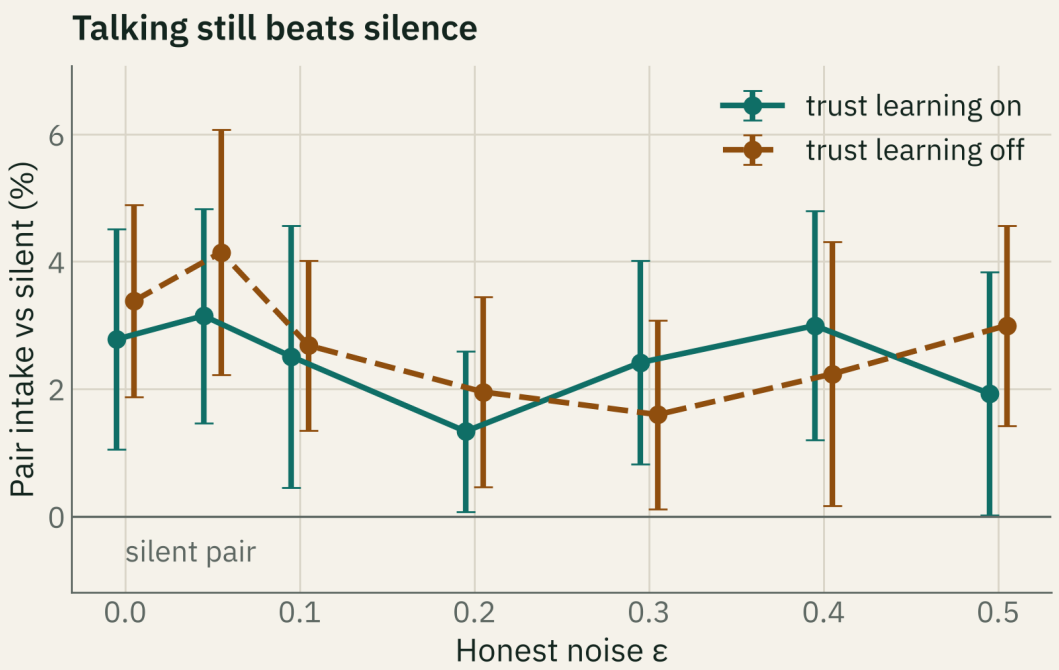
Dry

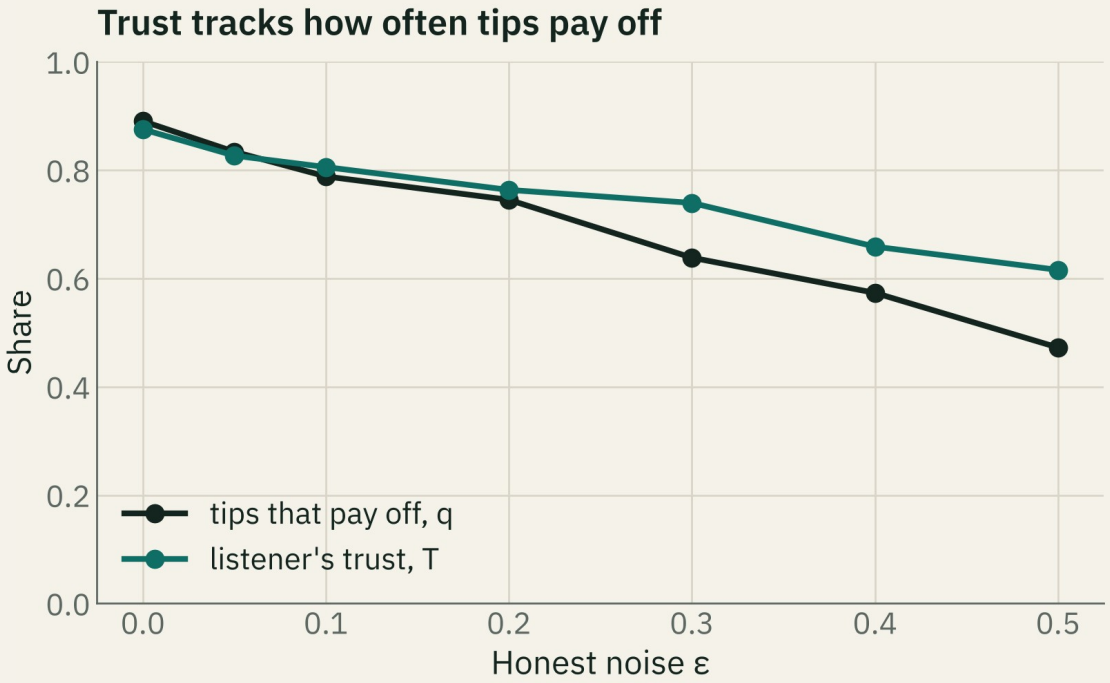
Adding the tree’s size doubles it, because small trees are not worth the walk.

Beyond size

Other trees and timing add nothing we can measure.

Even very noisy talk beats silence

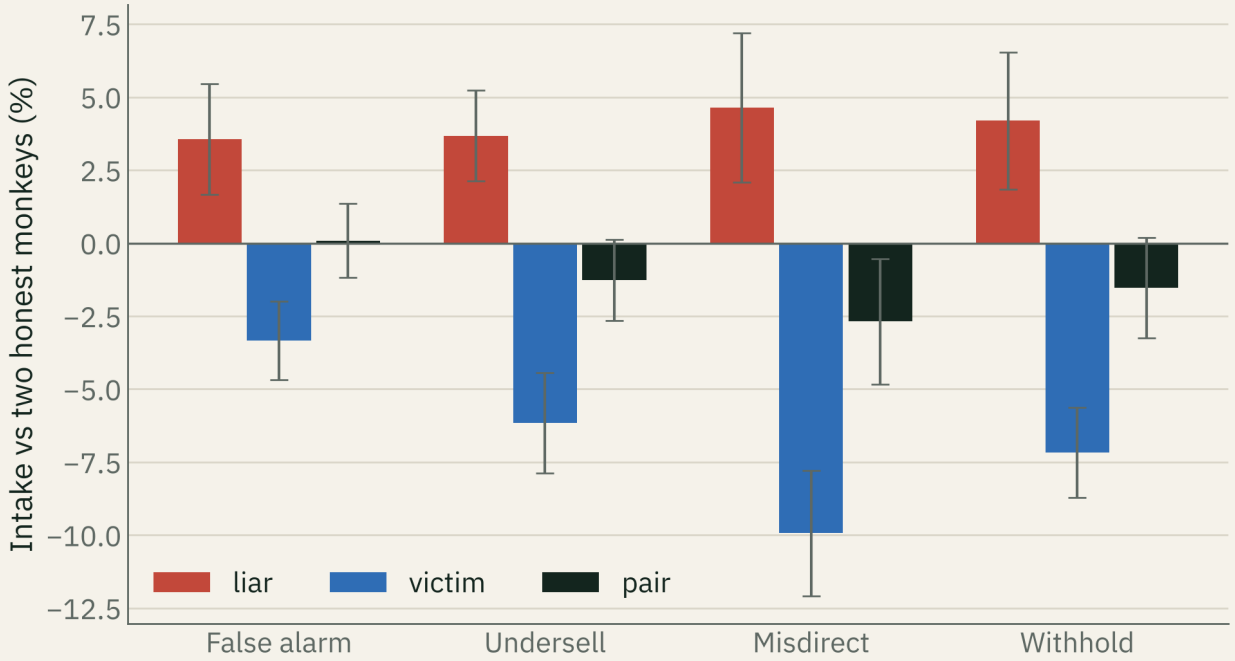


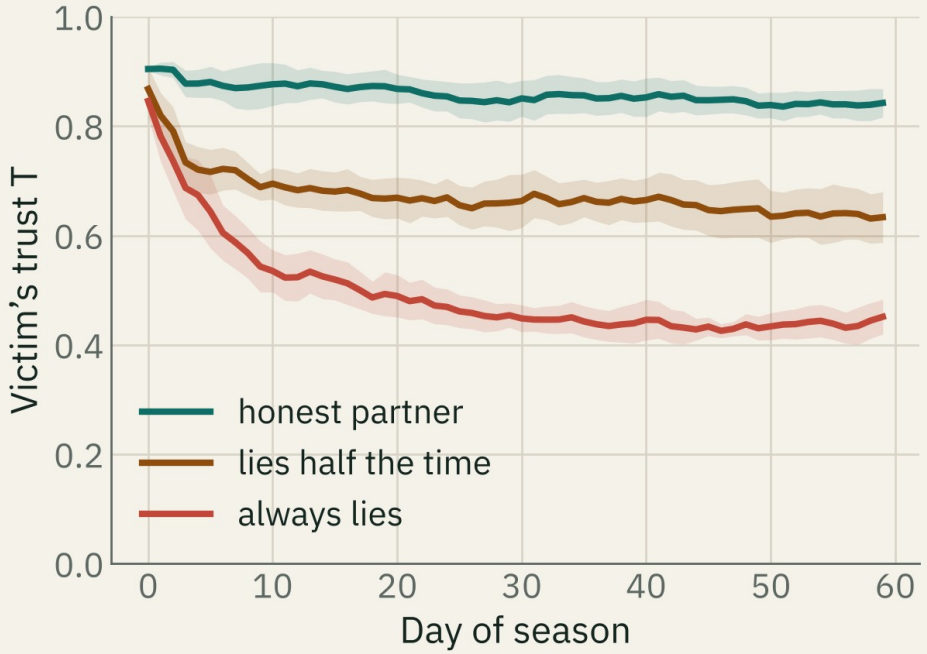


+2%

With half of every message garbled, only 47% of tips pay off, yet talking still adds about 2%. A listener walks only when a tip is worth it and checks it on arrival, so a bad tip costs one walk.

Lies move food from victim to liar





Liar

+4–5%

gains, with every tactic

Victim

−3 to −10%

loses

Pair

up to −3%

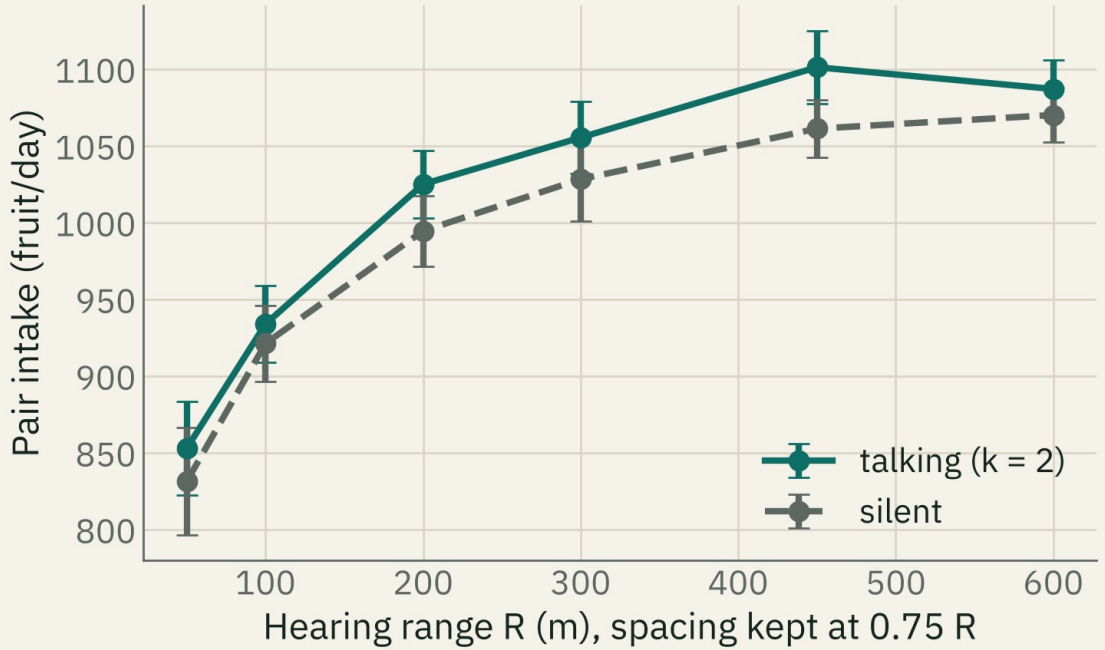
most from misdirection

Victim's trust vs. a constant liar

0.45

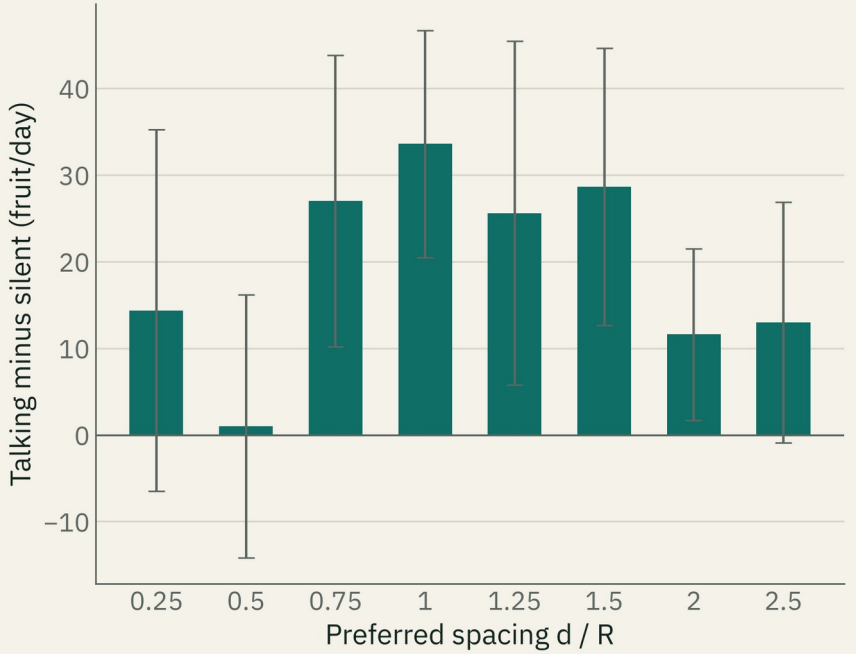
where our equation predicts, within ~20 days

Spreading out matters more than talking



Range and spacing

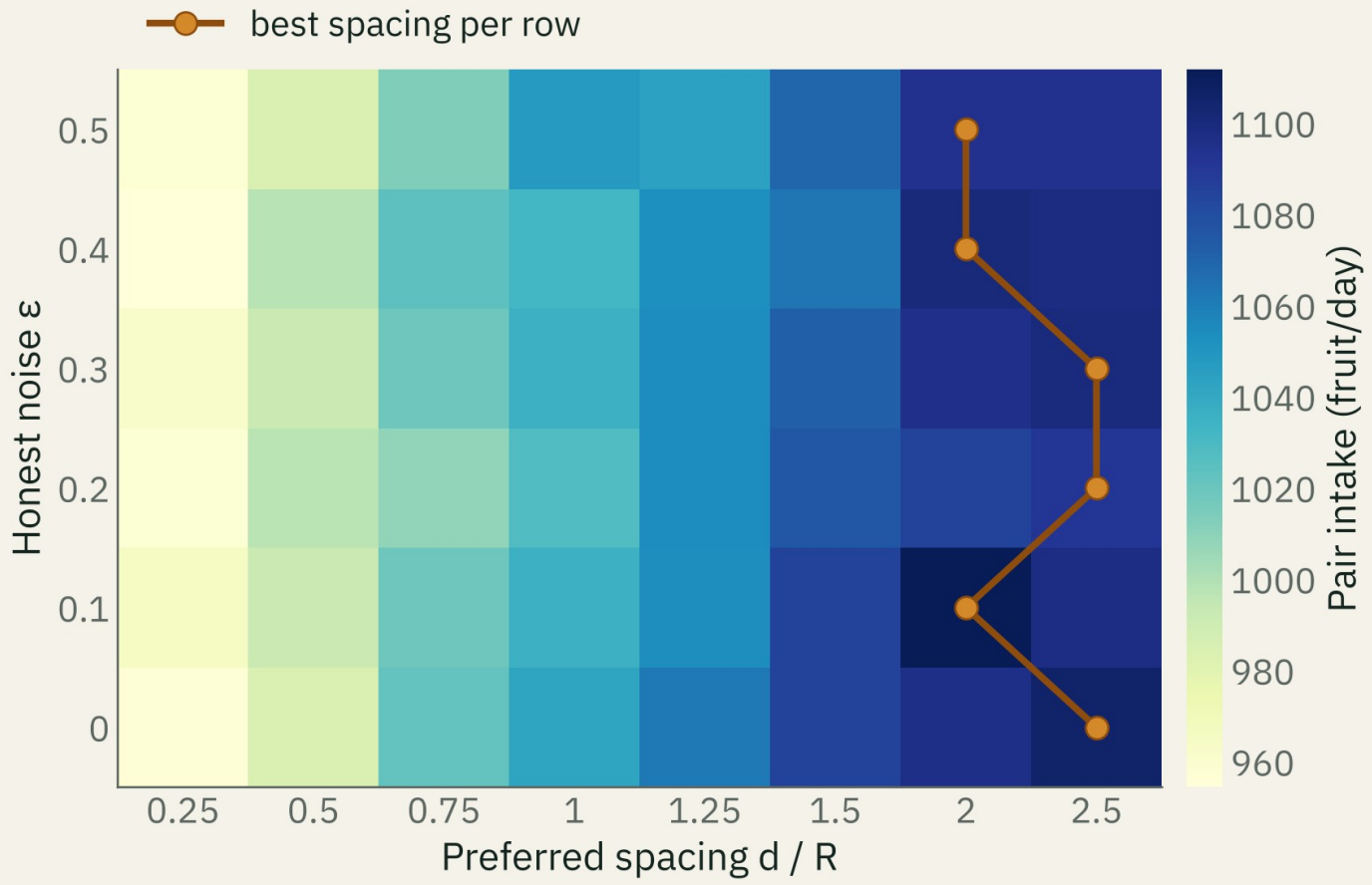
Longer range lets the pair spread out: silent pairs gain almost as much.



What talking adds

Talking pays most at 0.75–1.5 R, where partners are apart but still in earshot.

No trade-off: distance wins either way



Colour changes **across** the map, not up it.

15%

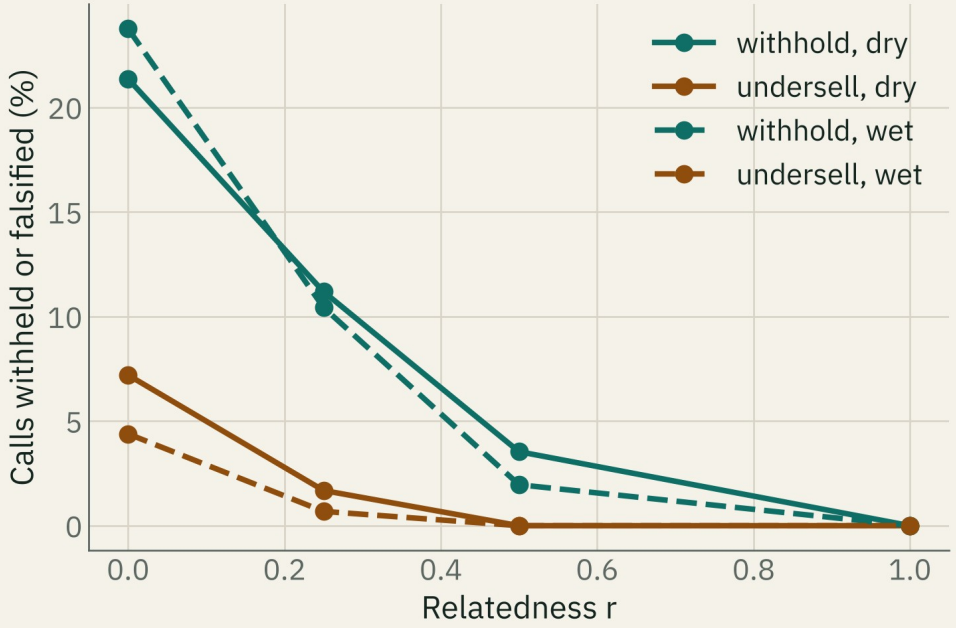
spacing moves intake by 15%; noise barely moves it.

The best spacing is 2–2.5 R at every noise level, and the same holds as honesty falls from 1 to 0.

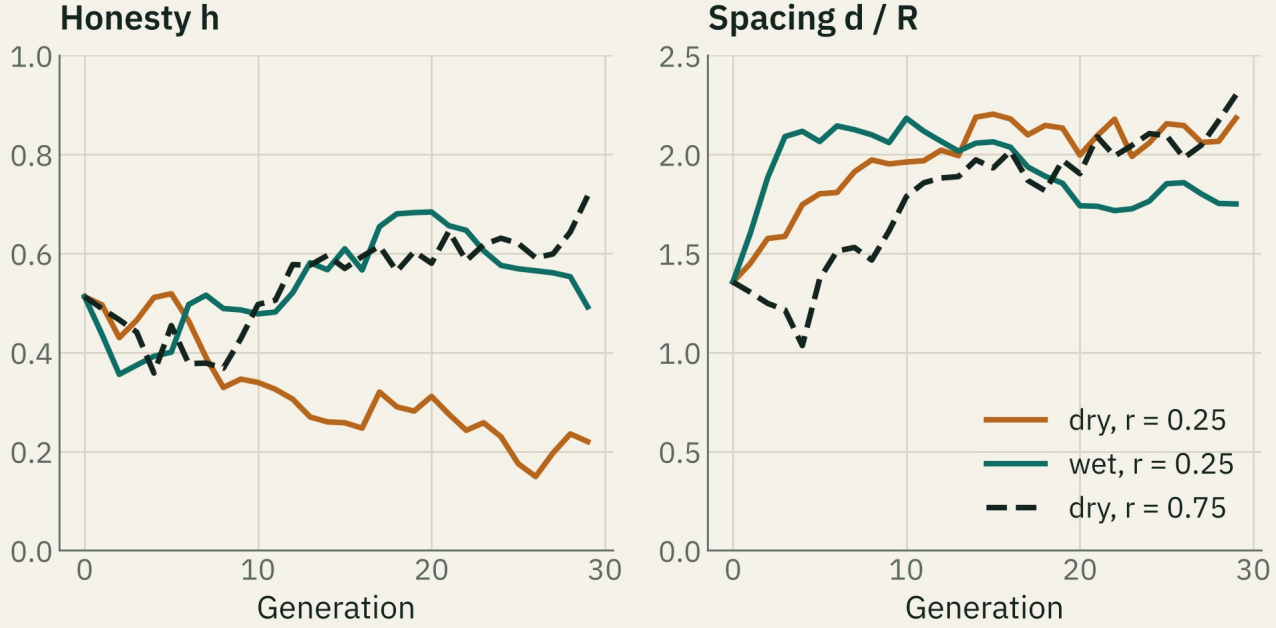
So there is no balance to strike: animals should spread out first and worry about accuracy second.

Kin stay honest; strangers stay quiet

Strategic monkeys: share of calls they cheat on



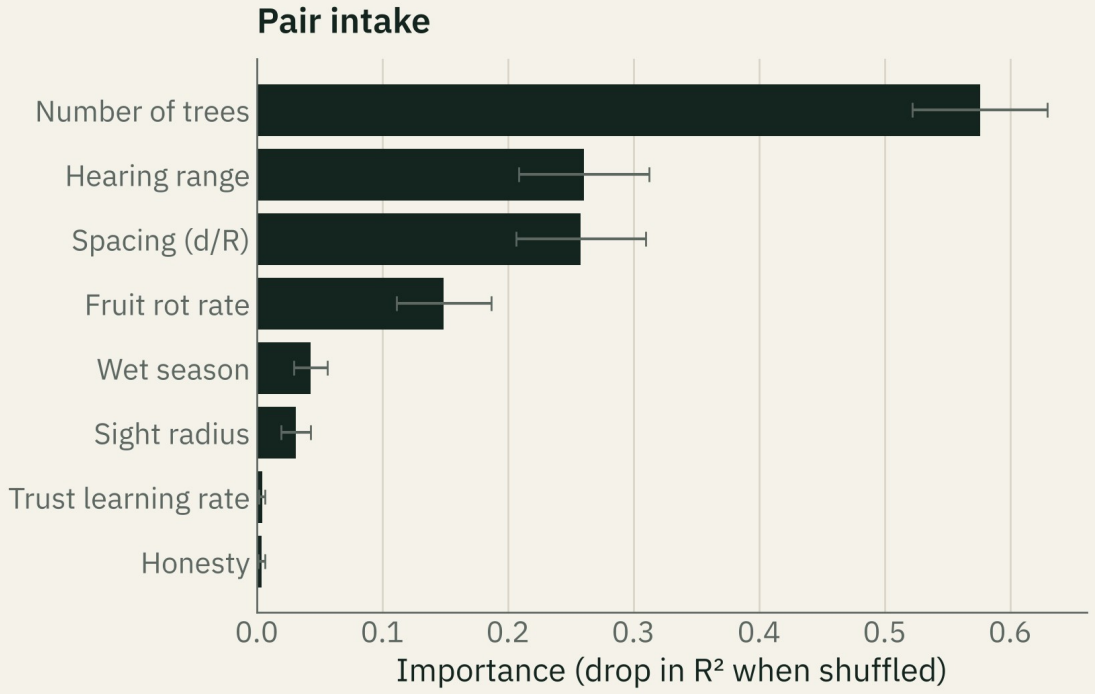
Evolving the strategy over 30 generations



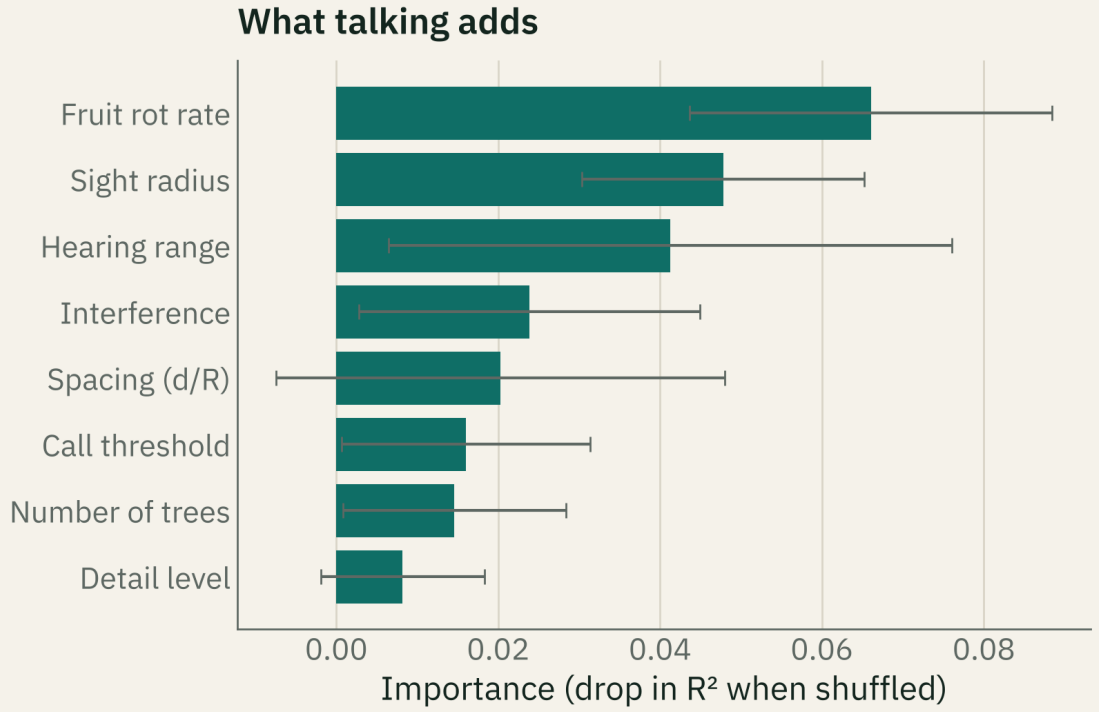
Cheaters withhold far more than they lie. Honesty survives with kinship ($h \approx 0.63$ at $r = 0.75$ vs 0.20 at $r = 0.25$), and every run evolves to spread out to about 2 R.

The forest first, then spacing and range

Total pair intake: driven by the forest, range and spacing



What talking adds: small and unpredictable



400 random parameter sets (Latin hypercube). Intake is driven by tree density, hearing range and spacing, which act as one ratio (H^2 = 0.37). What talking adds varies little and unpredictably: +1 fruit/day on average, positive in 43% of sets.

Three ODEs reproduce the simulation

Time shares: searching s, travelling to a known tree τ, feeding f

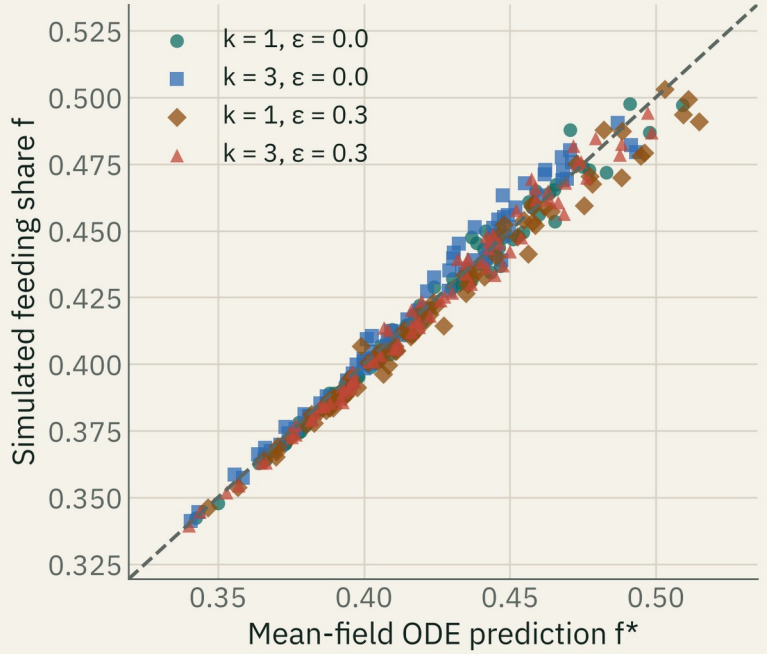
$$\frac{ds}{dt} = -(\lambda + \kappa)\, s + \mu(1 - \bar{q})\, \tau + \nu(1 - \varphi)\, f$$

$$\frac{d\tau}{dt} = (\lambda + \kappa)\, s + \nu\varphi f - \mu\tau$$

$$\frac{df}{dt} = \mu\bar{q}\, \tau - \nu f$$

$$\kappa = \rho_0\, p_{hear}(d)\, f \qquad \lambda(d) = \lambda_0\big(1 - b\, w(d)\big)$$

λ: finding food alone · κ: being called · μ: arriving · ν: leaving · q̄: tip pays off



Rates measured from each run; mean error 0.004 over 320 runs.

Threshold ρq/ν = 0.00–0.09, far below 1: calls alone cannot keep a pair fed, so talking adds only ~3%.

What matters, in order

1 **The forest**
How many trees fruit and how fast fruit rots set most of what a pair can eat.

2 **Spacing and hearing range**
Spreading out to about 2 hearing ranges adds 15%; it works as one ratio, d/R.

3 **Whether they talk**
+3%. Calls rarely reach a nearby, hungry partner ($pq/v < 0.1$).

4 **Detail and accuracy**
“Food here” plus tree size is enough; 50% noise still beats silence.

Distance vs accuracy

No trade-off. The best spacing stays at 2–2.5 R whether talk is perfect, noisy or dishonest.

Deception

Lies mostly move food from victim to liar (pair loses ≤ 3%). Cheaters prefer silence to lies; kinship keeps them honest.

Spacing and the forest set the harvest; talking is a fine-tuning factor, not a main driver.

What the model cannot yet tell us

Limits

Overlap is too high

Core ranges overlap 0.57–0.72 versus the paper's optimum of 1/3, because both monkeys share one sleeping site.

Only two animals

Real spider monkeys split and merge into changing subgroups; the paper's w* = 1/(n+1) depends on group size n.

Hand-set parameters

Rates come from the literature, not fitted to GPS tracks; dry-season monkeys visit 6 trees a day, not ~10.

Next steps

Fission–fusion

Groups of 3–10 with separate sleeping sites, to test whether w falls to 1/(n+1).

Fit to field data

Calibrate λ, μ, v against Punta Laguna tracks and fruit surveys.

Bigger runs

60 forests and 3,000 parameter sets (one command) to pin down what drives the talk gain.

Thank you

Questions?

Code: CANOPY (Python). Every figure regenerates with one command: **python run_all.py**

References

Ramos-Fernandez G. et al. (2026) Uncovering complementary information sharing in spider monkey collective foraging using higher-order spatial networks. *npj Complexity* 3, 1.

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Holling C.S. (1959) *Canadian Entomologist* 91 · Charnov E.L. (1976) Marginal value theorem. *Theor. Pop. Biol.* 9.

Crawford V. & Sobel J. (1982) Strategic information transmission. *Econometrica* 50 · Hamilton W.D. (1964) *J. Theor. Biol.* 7.

Taylor P.D. & Jonker L.B. (1978) *Math. Biosci.* 40 · Lundberg S.M. & Lee S.-I. (2017) SHAP, NeurIPS · Friedman J.H. & Popescu B.E. (2008) *Ann. Appl. Stat.* 2.

Where trust settles, solved by hand

$$\alpha_{n+1} = (1-\lambda)\,\alpha_n + \lambda\,\alpha_0 + o_n, \qquad T = \frac{\alpha}{\alpha+\beta}$$

β updates the same way with 1 − o. If a share *p* of tips check out, the expected step is $\alpha \leftarrow (1-\lambda)\alpha + \lambda\alpha_0 + p$.

Fixed point: $\alpha^* = \alpha_0 + p/\lambda$ and $\beta^* = \beta_0 + (1-p)/\lambda$, so

$$T^* = \frac{\alpha_0 + p/\lambda}{\alpha_0 + \beta_0 + 1/\lambda} = \frac{9 + 20p}{30}$$

Prior $\alpha_0 = 9$, $\beta_0 = 1$ ($T_0 = 0.9$); $\lambda = 0.05$ means trust remembers about the last $1/\lambda = 20$ checks.

Share of tips that check out, p	Trust settles at
1.0	0.97
0.75 (honest, noisy world)	0.80
0.5	0.63
0.25 (constant misdirector)	0.47
0	0.30

A strategic monkey lies only when it pays

$$\underbrace{\min(A, n_s) - \min(\tfrac{A}{2}, n_s)}_{\text{fruit I keep}} - r \underbrace{\min(\tfrac{A}{2}, 75)}_{\text{partner loses}} - \underbrace{30\,T}_{\text{reputation}} > 0$$

Fruit I keep

Zero when the tree is big enough for both: lying only tempts when food is short.

r × partner's loss

Relatedness r weighs kin's gain, as in Hamilton's rule.

Reputation

Proportional to the partner's trust. Zero for silence, which is never caught.

Can calls alone keep the pair fed?

Switch off solo discovery (λ = 0, φ = 0) and look near f = 0, where s ≈ 1. The travel and feeding equations become linear:

$$\frac{d}{dt}\begin{pmatrix}\tau\\ f\end{pmatrix} = \begin{pmatrix}-\mu & \rho_p\\ \mu q & -\nu\end{pmatrix}\begin{pmatrix}\tau\\ f\end{pmatrix}$$

Trace < 0, so f = 0 is unstable exactly when the determinant μv − μqρ_p is negative:

$$R_{info} = \frac{\rho_p\, q}{\nu} > 1$$

Read it like R_0

A feeding monkey recruits its partner at rate ρ_p for a bout of length 1/v; each recruitment pays off with chance q. Above 1, knowledge of food spreads on its own.

ρ_p = ρ₀ p_hear(d), so R_info falls as the pair spreads out or as accuracy q drops.

Where the numbers come from

Forest and body	Value	Source
Arena	800 m square	paper: ranges 20–70 ha
Trees per 30 days	140 dry · 250 wet	calibrated
Median crop	300 · 500	calibrated
Ripening window	3–8 · 12–30 days	assumed
Rot rate δ	0.3 per day	assumed
Search, travel speed	8, 20 m/min	fit to 1.75–3.3 km/day
Sight radius	25 m	paper: parties ≤ 50 m

Talk and strategy	Default	Varied over
Hearing range R	200 m	50–600 m
Spacing d_pref	0.75 R	0.25–2.5 R
Detail k	2	0–4
Honest noise ε	0.05	0–0.5
Honesty h	1	0–1
Prior trust, forgetting	0.9, λ = 0.05	λ: 0–0.5
Relatedness r	0.25	0–1

Holling II, the marginal value theorem and Lévy exponent 2 come from standard foraging theory. Experiment 8 varies 13 parameters at once to test which conclusions survive.